

Pretham Voruganti

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EDUCATION

Purdue University, West Lafayette, IN

Aug. 2023 – May 2026

Bachelor of Science in Cybersecurity

GPA: 3.06/4.0 currently

Relevant Coursework: Advanced Programming, Machine Learning & AI, Cloud Engineering, Network & System Administration, Information Security Mgmt

TECHNICAL KNOWLEDGE

Languages & Frameworks: Python, Java, C++, JavaScript, TypeScript, Swift, SQL, Bash, PowerShell, HTML

Cloud & DevOps: AWS (EC2, S3, Lambda, RDS, IAM), Terraform, Docker, Kubernetes, GitHub Actions, Firebase

Data Science & AI: Pandas, NumPy, Scikit-learn, TensorFlow, SciPy, Power BI, Data Visualization

Scientific Computing: Multiscale statistical analysis, covariance-aware χ^2 detection, null model simulation, point-cloud geometry

Security: Penetration Testing, Vulnerability Assessment, Scapy, NIDS, Kali Linux, CompTIA Security+, AWS AI/Cloud Practitioner

WORK EXPERIENCE

IT Security Analyst Intern | NSG Info, Alexandria, VA

May 2025 – Aug. 2025

- Maintained network security protocols and user access controls across enterprise infrastructure.
- Conducted vulnerability assessments and penetration testing to identify and remediate security gaps.
- Visualized and communicated findings to stakeholders using Power BI dashboards.

PROJECTS

Multiscale Clustering Detector — Cosmology & Molecular Biology

Jan. 2026 – Present

- Built a 3D point-cloud clustering detector using covariance-aware χ^2 statistics and logarithmic-scale null models, originally to test geometric structure hypotheses in galaxy distributions.
- Adapted the pipeline for Single-Molecule Localization Microscopy (SMLM): added blinking correction (union-find), cell-boundary null models, and nanometer-scale calibration.
- Designed a full validation suite (8 experiments): FPR calibration, power analysis, geometry robustness, and biological test cases modeled on synaptic receptors and nuclear pore complexes — outperforming DBSCAN and Ripley's K at low cluster fractions.

Network Intrusion Detection System

Nov. 2024 – May 2025

- Developed a custom NIDS in Python/Scapy with signature-based and anomaly-based detection techniques.
- Tested system performance against attack simulations using Kali Linux with a team of 4.

Smart Food Delivery Locker System

May 2024 – Present

- Built production backend with Express, TypeScript, Redis, and Knex for Bluetooth-enabled locker access and real-time delivery management.
- Designed predictive analytics pipelines in Python to forecast locker usage and optimize routing; implemented fault-tolerant job queues with BullMQ.

Analysis of Inflation — Data Science Project

Jan. 2024 – Apr. 2024

- Time-series analysis of BLS datasets using Pandas/NumPy; applied linear regression to model relationships between inflation, productivity, and key economic indicators.

SKILLS & ACTIVITIES

Capture the Flag (Cybersecurity): Cryptography, web exploitation, reverse engineering

Swift / Hackathons: iOS app development for competitions and personal projects